

Delhi Technological University (DTU)

**Assignment 1 - Data Warehouse and Data
Mining (DWDM)**

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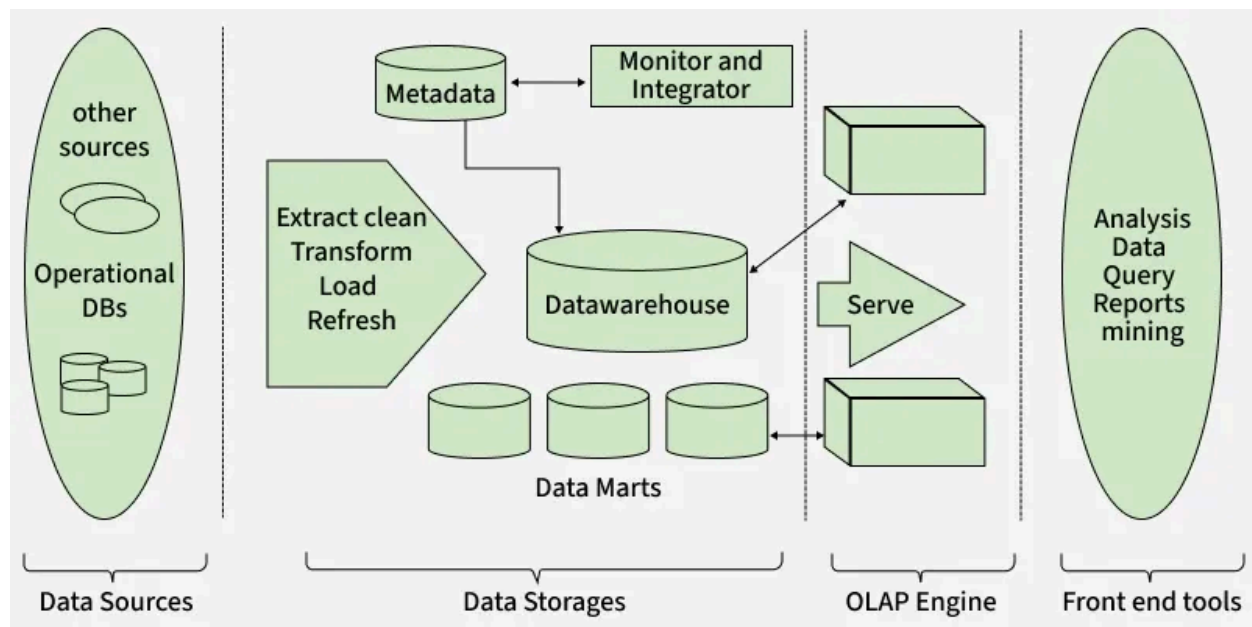
1. What is the KDD process? Explain its main steps and show how Data Mining fits into KDD.

KDD stands for **Knowledge Discovery in Databases**. It is the overall process of discovering useful knowledge from data. Data Mining is just one specific step within this larger process.

The main steps are:

1. **Data Selection:** Selecting the relevant data from different sources (databases, files) needed for analysis.
2. **Data Preprocessing:** Cleaning the data by handling missing values and removing noise (errors).
3. **Data Transformation:** Converting data into a format suitable for mining (e.g., normalizing values).
4. **Data Mining:** This is the core step where intelligent methods are applied to extract patterns.
5. **Interpretation/Evaluation:** Analyzing the found patterns to see if they are useful and understandable.

2. Draw and explain the 3-tier Data Warehouse Architecture with a neat diagram.



A Data Warehouse usually has three levels or "tiers":

- **Bottom Tier (Database Server):** This is the storage layer. It contains the data warehouse database where data is loaded after being extracted and cleaned from source systems (like transactional databases).

- **Middle Tier (OLAP Server):** This is the processing layer. It uses OLAP (Online Analytical Processing) servers to reorganize the data so it can be analyzed quickly. It handles multidimensional operations (like slicing and dicing data).
- **Top Tier (Client Layer):** This is the user interface. It consists of tools like query reporting, analysis tools, or data mining software that the user interacts with to view reports and graphs.

3. Distinguish between Nominal, Ordinal, Interval, and Ratio scales of measurement with suitable examples.

- **Nominal:** These are simple labels or categories with no specific order.
 - *Example:* Eye color (Blue, Brown, Green) or Gender (Male, Female).
- **Ordinal:** Categories that have a meaningful order or ranking, but the difference between them is not defined.
 - *Example:* T-shirt sizes (Small, Medium, Large) or Customer Satisfaction (Unhappy, Neutral, Happy).
- **Interval:** Ordered numbers where the difference between values is meaningful, but there is no "true zero" (zero doesn't mean "none").
 - *Example:* Temperature in Celsius (0°C is cold, not "no temperature") or Calendar years.
- **Ratio:** Similar to interval, but it has a "true zero," meaning you can say something is "twice as much" as something else.
 - *Example:* Weight (0 kg means no weight), Height, or Bank Balance.

4. Differentiate between OLTP and OLAP on at least four points with suitable examples.

3) OLTP vs OLAP (Expanded)

Item	OLTP	OLAP
Users	Clerks, IT professionals	Knowledge workers/analysts
Function	Day-to-day operations	Decision support
DB design	Application-oriented	Subject-oriented
Data	Current, up-to-date, detailed, flat, isolated	Historical, summarized, multidimensional, integrated
Usage pattern	Repetitive	Ad-hoc
Access	Read/Write; index/hash on primary key	Lots of scans
Unit of work	Short, simple transaction	Complex query
Records accessed	Tens	Millions
# of users	Thousands	Hundreds
DB size	100MB–GB	100GB–TB
Key metric	Transaction throughput	Query throughput, response time

5. Define Support and Confidence in Association Rule Mining with an example.

These metrics measure how strong a rule is (e.g., "If someone buys Bread, they will buy Butter").

- **Support:** Indicates how frequently the itemset appears in the dataset.
 - *Formula:* $(\text{Transactions containing both A and B}) / (\text{Total Transactions})$
- **Confidence:** Indicates how likely item B is purchased when item A is purchased.
 - *Formula:* $(\text{Transactions containing both A and B}) / (\text{Transactions containing A})$
- **Example:** Out of 100 transactions, 10 contain both Milk and Bread. 20 transactions contain just Milk.
 - Support for (Milk $\rightarrow$ Bread) = $10/100 = 10\%$.
 - Confidence for (Milk $\rightarrow$ Bread) = $10/20 = 50\%$.

6. Explain the key characteristics of a Data Warehouse (subject-oriented, integrated, time-variant, non-volatile) with examples.

1. **Subject-Oriented:** The data is organized around major subjects rather than applications.
 - *Example:* Instead of organizing by "Invoicing System," it organizes by "Customer," "Product," or "Sales."
2. **Integrated:** Data from different sources is made consistent (naming conventions, units).
 - *Example:* If one system uses "M/F" and another uses "1/0" for gender, the warehouse converts them all to "Male/Female."
3. **Time-Variant:** Data is stored as a series of snapshots over time, allowing historical analysis.
 - *Example:* You can see a customer's address from 5 years ago, whereas a standard operational database might only show their current address.
4. **Non-Volatile:** Data is mostly read-only. Once data is entered, it is not updated or deleted row-by-row like in a transaction system.
 - *Example:* You don't correct a spelling mistake in a 3-year-old sales record; you just load new data.

7. Compare the Apriori Algorithm and the FP-Growth Algorithm.

- **Apriori Algorithm:**
 - It uses a "candidate generation" approach (it guesses possible combinations and checks them).
 - It requires scanning the database multiple times (once for every size of itemset), which makes it slower on large datasets.
- **FP-Growth (Frequent Pattern Growth):**
 - It does *not* generate candidates. Instead, it builds a compact tree structure called an FP-Tree.
 - It only scans the database twice, making it much faster and more efficient than Apriori for large databases.

8. Give a brief explanation of Data Mining. Why is it important?

Data Mining is the process of analyzing large amounts of data to discover hidden patterns, correlations, and insights that typically cannot be found by simple queries.

Importance:

- **Decision Making:** Helps businesses make evidence-based decisions (e.g., what products to stock).
- **Fraud Detection:** Can identify unusual patterns in banking to stop credit card theft.
- **Marketing:** Helps companies target the right ads to the right people.

- **Scientific Discovery:** Helps in analyzing genetic data or medical records to find cures.

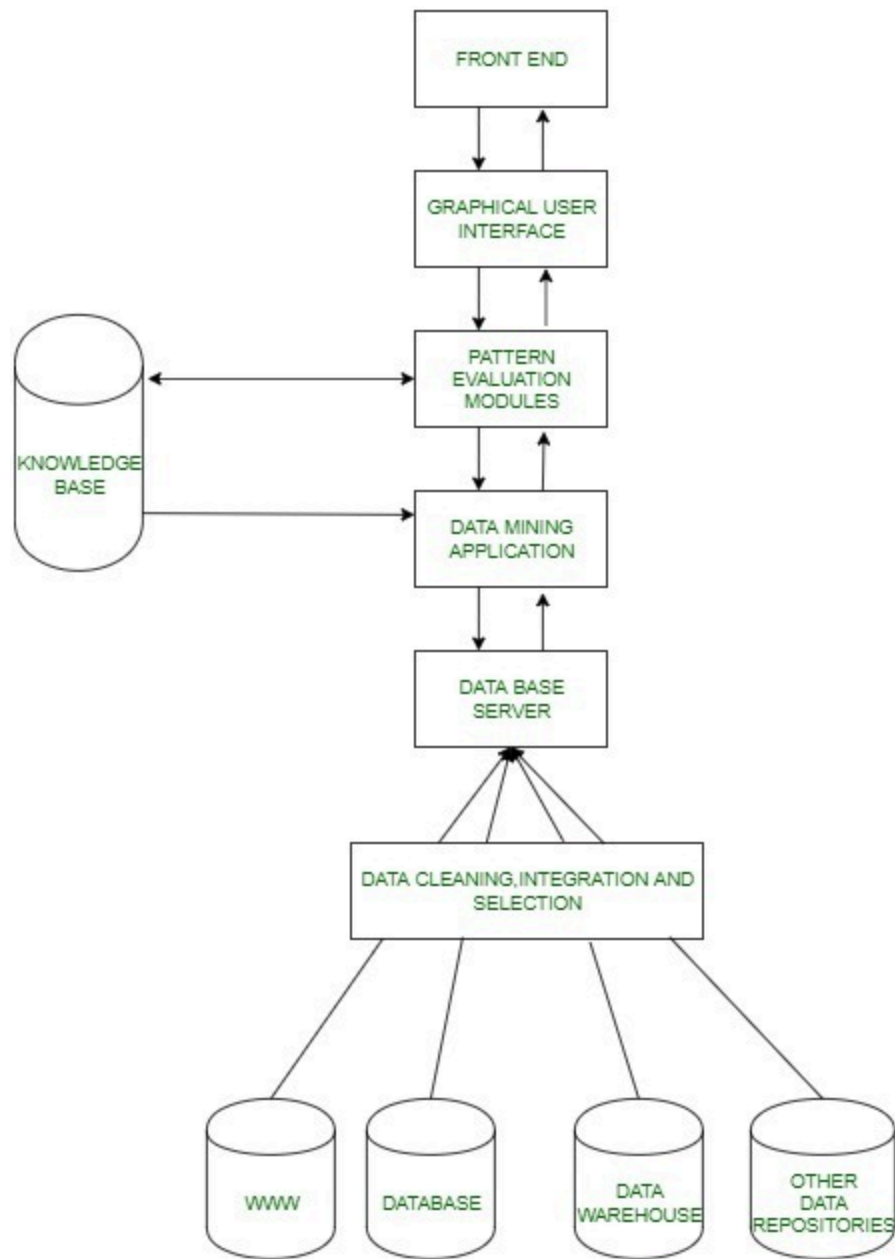
9. A retail chain wants to analyze its sales data using a data cube with dimensions Product, Location, and Time. Explain why computing a full cube may be inefficient. How can the concept of an iceberg cube help in this scenario? Illustrate with an example.

Computing a **full cube** involves calculating the totals for every possible combination of Product, Location, and Time. If you have thousands of products and locations, this creates a massive number of combinations, taking up huge memory and processing power. Most of these combinations might be empty (e.g., Winter Jackets sold in Summer in a specific small town).

Iceberg Cube: This is a strategy where we only compute and store the combinations that meet a certain minimum requirement (threshold), usually a minimum support count.

- **Example:** Instead of calculating sales for every product in every store on every day, the system only calculates and stores data for Product-Location combinations where sales were greater than \$10,000. We "throw away" the insignificant data (the bottom of the iceberg) to save resources.

10. Explain the Architectures of Data Mining Systems with a neat diagram.



A typical Data Mining System architecture consists of:

1. **Database/Data Warehouse:** The source where data is stored.
2. **Database/DW Server:** Fetches the relevant data based on the user's request.
3. **Knowledge Base:** Contains domain knowledge (like concept hierarchies) to guide the search.
4. **Data Mining Engine:** The core software that runs algorithms (clustering, classification, association) to find patterns.
5. **Pattern Evaluation Module:** Filters out uninteresting patterns and keeps only the useful ones.

6. **Graphical User Interface (GUI):** Visualizes the results for the user.

11. **With the help of a neat diagram, explain the concept of Association Rule Mining. Also define support and confidence.**

Association Rule Mining is a technique used to find relationships between variables in large databases. It is most commonly used for "Market Basket Analysis" to see which items are bought together.

- *Note:* The definitions for **Support** and **Confidence** are provided in Answer #5. You can refer to those definitions here.

12. **What are multi-dimensional association rules? Differentiate between inter-dimensional, hybrid-dimensional, and quantitative rules with suitable examples.**

Multi-dimensional rules involve more than just "items" bought; they include other attributes like age, income, or location.

1. **Inter-dimensional Rules:** No predicate (attribute) is repeated.
 - *Example:* Age(20-25) AND Occupation(Student) implies Buys(Laptop). (Different attributes: Age, Occupation, Buys).
2. **Hybrid-dimensional Rules:** A predicate is repeated in the rule.
 - *Example:* Age(20-25) AND Buys(Laptop) implies Buys(Mouse). (The attribute "Buys" appears twice).
3. **Quantitative Rules:** Rules involving numeric attributes that are discretized (broken into ranges).
 - *Example:* Age(30-35) AND Income(50k-70k) implies Buys(Car).

13. **What is the role of Metadata in a Data Warehouse? Explain with examples.**

Metadata is "data about data." It acts as a directory or a roadmap for the Data Warehouse.

Roles/Examples:

- **Technical Metadata:** Describes the structure.
 - *Example:* Table names, column names, data types (Integer, Varchar), and how data maps from the source to the warehouse.
- **Business Metadata:** Describes the meaning for users.
 - *Example:* Definitions of business terms (e.g., "What exactly is 'Gross Profit'?"), currency units used, and ownership of data.

14. List and explain any four applications of Data Mining in real life.

1. **Retail/Marketing:** Market Basket Analysis helps stores understand that people who buy diapers also buy beer (classic example), allowing for better store layout and promotions.
2. **Banking/Finance:** Credit scoring uses data mining to predict if a customer will default on a loan based on their history.
3. **Telecommunications:** Churn prediction helps phone companies identify customers who are about to cancel their subscription so they can offer them a deal to stay.
4. **Healthcare:** Analyzing patient records to associate symptoms with diseases or to predict outbreaks of illnesses in specific regions.

15. Why can support and confidence measures sometimes lead to misleading association rules? How does correlation (lift) improve rule evaluation?

Support and Confidence can be misleading if the item in the "consequence" is already very popular.

- **The Problem:** Suppose 90% of all customers buy Bread. If you find a rule `Buy(Apples) -> Buy(Bread)` with 85% confidence, it looks strong. But actually, buying Apples *lowered* the probability of buying Bread (from 90% down to 85%). The rule is misleading.
- **The Solution (Lift):** Lift measures correlation. It compares the confidence of the rule to the expected confidence if the two items were independent.
 - **Lift > 1:** Positive correlation (They help each other).
 - **Lift = 1:** No correlation (Independent).
 - **Lift < 1:** Negative correlation (They discourage each other).
 - Lift helps identify rules that represent a *real* relationship, not just a coincidence caused by high popularity.